

Chapter 11

Motivation in the College Classroom

Few topics concern teachers at all levels as much as the motivation of students. We worry over the students who appear disengaged, who attend sporadically, or whose laptop use in class seems more directed toward social networking than note taking, and we may disparage those who appear to care only about grades. We delight in the students who share our passion for the subject matter, who are eager to ask intelligent questions, who view grades as informational feedback, and who not only prepare for class but seek us out to learn more. We marvel when we compare notes with a colleague and learn that these contrasting motivational profiles sometimes describe the same student—but in different courses, suggesting that motivation is something other than an abiding characteristic of an individual.

We all want students who are motivated to learn. These are the students who choose to attend class regularly, participate constructively, persist when learning is difficult, make the effort to prepare for class and to study effectively, who solicit help when they need it, and who translate all this into academic success. Knowing more about how students are motivated, and what you can do to structure a class that positively affects student motivation, can make a significant difference in student engagement and learning. A classroom of motivated learners affects *our* motivation

This chapter was written by Barbara Hofer of Middlebury College.

as well and can make teaching a more satisfying experience for the instructor. Particularly in an era where we are competing for student attention among so many technological and communication devices, knowing more about classroom motivation is increasingly relevant.

MOTIVATIONAL THEORIES: AN OVERVIEW

Researchers typically consider three indices of motivation: choice, effort, and persistence; achievement is an outcome of these variables. Accordingly, students who are motivated to learn choose tasks that enhance their learning, work hard at those tasks, and persist in the face of difficulty in order to attain their goals. So it should be no surprise that motivation is important to consider if we want to enhance student learning. Why students vary in their motivation is a compelling question, and several theoretical frameworks help provide answers.

Some students may be driven by a high *need for achievement* (McClelland et al., 1953). Need for achievement may be characterized as an individual trait or disposition, and it is likely to be the outcome of early environments in which parents set high standards and valued achievement. In general, students differ from one another in the degree to which achievement for its own sake is meaningful to them, but this difference explains only one aspect of motivation, which is considered to be contextual and malleable, and a function of classroom instruction, tasks, and activities (Maehr and Zusho, 2009) as well as interest. A particular student may exhibit a striving for achievement on the soccer field but not in your class, or perhaps appears more motivated to achieve in some parts of your course than in others; and we have all known students who did not appear motivated at the start of a course but became deeply engaged. Most importantly, classroom environment and instructional practices can foster certain types of motivation over others, as can the overall climate of an educational institution.

Autonomy and Self-Determination

Many psychologists posit that human beings have a fundamental need for autonomy and self-determination (Deci and Ryan, 2000). In general, individuals want to be in charge of their own behavior, and they value a sense of control over their environment. We can enhance students' sense of control by offering meaningful opportunities for choice and by supporting their autonomy, which in turn enhances motivation. Quite often these opportunities for choice can be relatively simple things such as a choice of paper topics, test questions, due dates, or reading assignments,

yet they go a long way toward acknowledging a student perspective. For example, I typically list on my syllabus three due dates for papers and tell students that they can choose which of these dates work best for them. The actual assignments are similar at each of these points, but draw on material from that section of the course, and I list four or five topics from which students can choose. Too much choice, however, is not helpful, as Barry Schwartz (2004) has shown in *The Paradox of Choice*. Thus students typically don't benefit as much from highly open-ended assignments such as being allowed to "write about anything" they choose. Providing some parameters and guidance, and some choice within that framework, seems most effective (and simplifies your assessment).

Intrinsic and Extrinsic Motivation

Most educational researchers acknowledge that what matters is not only whether a student is motivated but also what type of motivation the student has. Instructors at the college level often complain of student preoccupation with grades, typified by the perpetual classroom question "Will that be on the test?" *Extrinsically* motivated students are likely to engage in the course for reasons of external rewards, such as grades, recognition, or the approval of others (notably instructors and parents). Individuals who are *intrinsically* motivated engage in an activity for the value of the activity itself, rather than for an external reward. These students learn for the pleasure of learning and have a sense of self-determination about their educational path. Intrinsic motivation has been shown to foster conceptual understanding, creativity, involvement, and a preference for challenge. Research on college-student learning indicates that students with an intrinsic orientation are more likely to use cognitive strategies such as elaboration and organization, resulting in deeper processing of the material (Pintrich and Garcia, 1991).

Although the image of a classroom of intrinsically motivated learners might sound ideal, students are also driven by the desire for grades, approval, praise, and other rewards, and understandably so. Intrinsic motivation and extrinsic motivation exist not on a single continuum but on two separate ones, and students may often have multiple goals for the same course. A student enrolled in a required course, for example, may be deeply interested in the material and may also see it as a step in her professional development; she may desire to earn an A so that she will be likely to gain admission to graduate or professional school. Even students who initially appear only extrinsically motivated to take a course, perhaps viewing it merely as a requirement toward graduation, can become more intrinsically motivated if the instructor arouses their curiosity, provides appropriate levels of challenge, and offers them choices

that enhance their control (Lepper and Hodell, 1989). Faculty members are also excellent models for intrinsic motivation, and talking about your own enthusiasm and passion for the field can be contagious. Motivation that begins as extrinsic can become internalized over time and we can help students come to an appreciation of learning that is not so driven by external rewards and approval.

Although studies have indicated that external rewards may diminish intrinsic motivation by undermining self-determination (Ryan and Deci, 2000), recent research seems to support the judicious use of external rewards as a complement to other motivational approaches. Extrinsic rewards may be particularly useful when intrinsic motivation is lacking—and it is reasonable to assume that students are not always going to be intrinsically motivated to learn everything they are expected to learn during the college years. Students may also find extrinsic rewards to be productive during the early stages of learning a new subject before they feel they can begin to master it and when the necessary nature of the tasks (such as memorizing vocabulary in a foreign language or learning a large number of terms in the sciences) may not be intrinsically interesting. There is also evidence from a study of intrinsic motivation in college undergraduates that the pursuit of grades may not be all bad (Covington, 1999), in that the attainment of grade goals can foster an increase in interest, at least among those whose goals are not driven by the desire to avoid failure.

Extrinsic rewards are most beneficial when they *contain informative feedback* and enable students to focus on improvement. Thus grades alone are less helpful than grades accompanied by narrative feedback that addresses specific directions for change. It is also possible to defer awarding grades and provide comments for improvement only. In the required Research Methods course that I teach, I give feedback—but no grades—on the first draft of research proposals, conveying how students might improve their work in the next version. With no grade to focus on, they seem to attend more deeply to the comments. Helping students separate feedback and grades, as well as providing opportunities for revisions, can help them tune more to improvement.

Taking the time to provide students with constructive feedback on papers as well as using class time when returning tests as an opportunity for further teaching can facilitate student engagement and motivation. Students appreciate knowing that their learning matters to you, and that you can help guide them toward success. I typically do a frequency count of missed items on a test, for example, and I use a few minutes of class to review any questions that more than a few students missed so that I can address misunderstandings. If more than half of the class misses an item, I don't count it and try to determine if it was simply a poorly written question or if I need to spend more time re-teaching the material.

Expectancy-Value Theory

Students typically direct their behavior toward activities that they value and in which they have some expectancy of success (Wigfield and Eccles, 2000). From this social cognitive perspective, motivation is viewed as the outcome of multiplying these two forces; if either one is absent, the resulting product is zero. Instructors can benefit by knowing that they need to foster both. Students need to feel that there is a reasonable possibility of success and that the work is of value. Thus even students who believe they can do well in an introductory course might not continue with the subject if they do not see that learning the material is worthwhile; likewise, even those who entered with professional ambitions dependent on the course may not persist in the field if they think that they cannot expect success. You may assume that students know the value of your field or of your particular course, but often this is not the case, and it may be worth the time to explain the relevance of what you are teaching. Fostering expectancy for success is equally important. Students benefit when instructors have high expectations for success and also provide the conditions for achieving it. Informative feedback is also critical here, as is availability in office hours or providing access to tutors, whether graduate student assistants or undergraduate peers.

Mastery and Performance Goals

Motivated behavior is directed toward goals, and goals related to learning tend to reflect two broad types of purposes: mastery goals and performance goals (Ames, 1992). Students who adopt *mastery goals* are those whose primary desire is to understand and master the material, with the goal of developing competence. By contrast, students with *performance goals* are more likely to focus on their achievement relative to the performance of others, directed more toward the demonstration of competence rather than its development. The classrooms we create may implicitly foster either type of goals (Meece, Anderman, and Anderman, 2006), depending on grading practices, classroom climate, and other such factors. Researchers have increasingly demonstrated the relevance of this model to higher education Zusho, A., Karabenick, S., Sims, B.C., and Rhee, C.K. (2007), and faculty members can be particularly influential in affecting productive motivational beliefs.

In a class that is focused on mastery, instructors generally use criterion-referenced grading rather than normative (grading on a curve), foster a supportive climate where students can take intellectual risks, and provide opportunities for students to demonstrate improvement. A mastery orientation may be visible in classroom discussions when students ask

genuine questions to which they do not already know the answers, driven by a desire to better understand the material, rather than to impress their peers and the instructor. Mistakes are viewed as an opportunity for learning, rather than as a measure of worth, subject to the judgment of others.

In a class that is focused on performance, instructors often use normative grading practices, which imply that only a specified percentage of students are likely to succeed no matter how hard the entire class works, and provide no opportunities for revising and improving written work. Student questions may be formulated to present the inquirer in the best light and to gain recognition and reward. In contrast to students with mastery goals, students who are ego-involved with their performance may compare grades with one another and take academic shortcuts, such as avoiding more effort than is necessary to acquire the desired grade or engaging in academic dishonesty (Jordan, 2001).

Overall, mastery goals lead to more adaptive outcomes, for such students are likely to focus on learning, use effective cognitive strategies, and experience less performance-impeding anxiety (Pintrich, 2003). Students in highly competitive college classrooms, however—which are performance oriented by design—may find it adaptive to pursue a performance orientation (Harackiewicz, Barron, and Elliott, 1998). Faculty members thus may have considerable power in shaping goal approaches within their classrooms. Fostering a particular goal orientation begins with course design and syllabus construction, when we make choices about evaluation and grading practices and how we plan to communicate them to students. Take time the first day of class to explain your grading practices and, if you are using a criterion-referenced grading system, to make it clear that students are not competing with one another for grades. This also makes it evident that it is advantageous to form study groups and support one another's learning. Students in required courses that are essential for their progression toward professional goals (medical school, business school, etc.) can benefit from an environment that teaches them that their success does not come at the expense of others.

Goal orientation is also reflected in an array of teaching practices. Mastery orientation thrives in a classroom climate of warmth and acceptance where instructors support and value intellectual risk-taking and avoid comparisons among students. Research indicates that student perceptions of classroom goal structures matter. For example, college students who view their instructors as promoting a mastery orientation, and whose instructors appear caring and supportive of questioning, show higher academic achievement in those courses. These perceptions of mastery goals have been linked to positive outcomes in a range of studies, including a preference for challenging tasks, enjoyment of learning, help-seeking, and adaptive learning strategies (Maehr and Zusho, 2009).

When professors foster a performance orientation by stressing competition and acknowledging only high-achieving students, students are less likely to exhibit such outcomes.

Social Motivation and Other Goals

Students are obviously motivated by more than academic achievement, and the focus on mastery and performance may overlook other goals students have (Zusho and Clayton, 2011). For example, they also have social goals that are operative in the classroom: they want to be socially responsible and to form social relationships with peers (Patrick, Hicks, and Ryan, 1997; Wentzel and Wigfield, 1998). Culture also shapes goals, and social goals may be particularly relevant to students from more collectivist cultures, where relational and group-oriented goals are common, rather than the individualistic ones suggested by the mastery and performance research (King and Watkins, 2011). (See chapter 12 for more information on teaching culturally diverse students.)

Although most studies of the relation between social goals and academic motivation and achievement have been conducted with younger adolescents, certainly no college instructor would doubt that social goals are operative in the college classroom. Enabling students to make new acquaintances in your classroom in conjunction with meeting academic goals may enhance student motivation to attend class and to participate in academic work. For example, a brief moment to discuss a question with a partner works well from a cognitive perspective because it fosters elaboration and retention and provides opportunities for clarification, but it also gives students an opportunity to get to know one another. Helping students form study groups prior to exams fosters preparation and also addresses social needs. Group projects, as long as they are well structured and include individual accountability, can also be ways for students to meet new people and make friends with common interests.

Social goals are not met solely in the classroom, nor are many of the other goals students might have. Given the various demands on their time, it may be no surprise that in spite of the best efforts of instructors, students are often weighing the benefits of how deeply to invest in a course or assignment. What may appear to instructors as an aversion to work or a disappointing level of effort, may be the result of a pragmatic decision on the part of students, as they balance an array of demands. Helping them learn how to be strategic in their choices in a way that does not hinder their academic work can be especially effective (e.g., when to read deeply and when to skim). Again, allowing choices of due dates in the syllabus also helps, as it offers students more control about the timing of tasks and helps alleviate some of the pressure they may feel.

Attribution Theory

When individuals need to seek an explanation for unexpected outcomes, they make attributions about the probable causes, and these attributions have motivational consequences (Weiner, 2001). In the academic sphere, this often arises when students fail to perform well on a test or get a grade that differs from what they had expected. Typical attributions are effort ("I didn't study hard enough"), ability ("I'm just not good at this subject"), or luck ("The test emphasized the material I actually studied!"). Attributions can be categorized along three dimensions: locus, stability, and responsibility, which refer respectively to whether the cause is internal or external, stable or unstable, and whether the cause is controllable or not. Students who explain their disappointments with internal, controllable attributions ("I know I didn't prepare adequately for the test") are likely to do better next time, because they believe they can affect the outcome. Students who attribute failure to stable, uncontrollable causes ("I will never understand statistics") are less likely to be motivated for improvement and understandably pessimistic about future outcomes.

Instructors can assist students in making adaptive attributions, particularly by helping them attribute failure to effort rather than ability, as well as by communicating their own positive attributions about students' capabilities to learn. When meeting with students to confer about low performance or an unexpected poor grade, you can help them reframe their thinking about the cause of their difficulties and help them gain a sense of control over future outcomes by helping them think diagnostically and rationally. Ask them to describe how they studied (or went about writing the paper), review the types of questions they missed or the most significant flaws in a paper, and help them know how to prepare or write more effectively in the future. In addition, referring students to a study skills center on your campus in order to improve their learning strategies can communicate that the problem is remediable and that they can take charge in addressing it. Overall, helping students endorse controllable attributions as explanations for failures can lead to increased performance, as attribution retraining with first-year students shows (Perry et al., 2010).

The Motivational Power of Beliefs about Intelligence

Fostering students' beliefs about the power of effort and hard work also reinforces the notion that intelligence is incremental and can be developed, a powerful motivational belief, according to Carol Dweck, author of *Mindset* (2006). Students with this "growth mindset" are likely to take

on challenging work, learn from mistakes, and approach new learning with enthusiasm, as these are opportunities for improvement. By contrast, individuals who believe in an entity view of intelligence, or what she calls a fixed mindset, are dedicated to preserving their own and others' impressions of their ability. Such students fear that having to work hard for accomplishment simply displays a lack of ability and talent, so even if they do exert effort they may try to hide it. Those with a growth mindset are more likely to appreciate how they have developed in the process of applying themselves well.

Dweck's research, supported by a large number of experimental studies, shows that the feedback of teachers (and parents and coaches and employers) has dramatic effects on how students perceive their own intelligence and what mindset they adopt. A student who does well on a paper and is told, "You're really very smart at this," will do what it takes to protect that image, perhaps not speaking in class unless they are confident of the answer, choosing less demanding assignments, and avoiding novel tasks. A student who is told, "You did well at this and must have worked hard to reach such a level of understanding," is likely to feel acknowledged for the effort and want to continue to seek challenges.

PUTTING MOTIVATION THEORY INTO PRACTICE

These principles can be used in many ways to structure classes that foster student motivation to learn. Here are a few suggestions:

1. When planning assignments, consider issues of choice and control. If you would like students to write two papers during the term, provide assignments during three time periods and let them choose which two to complete. This enables students to take charge of planning their work in the context of requirements from other courses and allows them to select issues of greatest interest. (This also has the advantage of spreading out the grading that you will need to do, an added bonus.) Similarly, provide a choice of topics for each assignment and consider a range of options that engage interest. Foster initiative by allowing students to propose alternative topics that meet the intent of the assignment.
2. Project your own motivation—for the subject matter and for the students. Take opportunities to describe your own intrinsic motivation for both research and teaching and your mastery orientation to learning. Too much of the literature on faculty "rewards" has focused on the extrinsic reinforcement for teaching, neglecting our own intrinsic motivation for academic work (as well as the intrinsic satisfaction of teaching). You are a powerful role model for your students as they develop their own passion

and motivation for learning as well as for their future professions. Get to know your students as individuals with lives beyond your classroom.

3. Foster students' intrinsic motivation to attend class by being well prepared, making lectures and discussions interesting, varying the instructional format, inducing cognitive dissonance and stimulating thought, and adding interactive elements where appropriate. Students are more motivated to come to class when the learning experience clearly exceeds what can be copied from another's notes. Engagement is more than being present, of course; they are also less likely to be texting under the desk or viewing Facebook rather than taking notes if the material is presented in an engaging way and supplements the reading they can do outside of class.

4. Make the value of your courses explicit, and take time to help students understand why what they are learning matters. Teach with a sense of purpose.

5. Promote adaptive attributions, helping students value the application of effort and learning strategies, and communicate your belief in their capability. Avoid praising students for ability or talent.

6. Foster mastery by encouraging students to revise their writing. Although it might not be reasonable for you to read drafts of every paper, you might do this for the first written assignment and then create peer review groups for additional papers. Or you can vary this process by responding to outlines for one paper and then reading drafts of opening paragraphs for the second. You can further foster mastery by uncoupling feedback and grading, so that early drafts receive written comments but no grades.

7. Adopt a criterion-referenced approach to grading rather than a normative one. Outline course requirements so that the point value for each assignment is clear from the beginning, and students know what they need to do to succeed—and know that they can succeed without worrying about their standing relative to others in the course. This fosters a sense of control, creates a cooperative rather than a competitive climate, and appeals to both intrinsically and extrinsically motivated students.

8. Test frequently enough that students become accustomed to the format and have opportunities to learn from their mistakes; at the very least, consider a similar format for the midterm and final. Allow students to justify and elaborate on their multiple-choice answers (which enhances control), and give partial or full credit for acceptable and reasonable justifications of alternative answers. Provide choices of essay questions to answer (e.g., "Answer five of the following six questions"). Consider providing one of the essay questions in advance, particularly one that might require more thoughtfulness and preparation.

9. When grading tests, consider dropping questions missed by a large number of students—and then re-teach the material when you return the tests. This sense of shared responsibility for the learning process heightens student awareness that you are committed more to their mastery of the material than to penalizing them for what they do not yet know (or for exam questions that might have been confusing).

10. Provide feedback that is constructive, noncontrolling (e.g., avoid words like "should" and "must"), and informative, thus enhancing student desire to improve and to continue to learn. View problems as something that can be addressed, not as evidence of an individual's worth.

If you are at a large university and involved in supervising teaching assistants, you can also make the motivational implications of your instructional decisions explicit to them. I am indebted to Paul Pintrich, Bill McKeachie, and Scott Paris, who were extraordinary role models in their design of graduate seminars that fostered student motivation, but who also provided me with opportunities as a teaching assistant to understand the motivational structure of their undergraduate courses, which I have enjoyed putting into practice in my own teaching.

Supplementary Reading

Although some of the following works are directed more toward the motivational issues of K–12 schooling, the theories and many of the suggestions are useful to those who are interested in the issue of motivation in the college classroom.

- ▶ J. Brophy, *Motivating Students to Learn* (Mahwah, NJ: Erlbaum, 2010). Writing for teachers, Brophy distills the motivation research into key principles that guide suggested classroom strategies.
- ▶ C. Dweck, *Mindset, The New Psychology of Success* (New York: Ballantine, 2007). This book makes the research on beliefs about intelligence highly accessible to a wide audience and gives examples for teaching, coaching, and parenting.
- ▶ D. H. Schunk, P. R. Pintrich, and J. Meece, *Motivation in Education: Theory, Research, and Applications*, 3rd ed. (Upper Saddle River, NJ: Merrill/Prentice Hall, 2007). This primer for instructors on a wide variety of motivational theories and their application surveys the research thoroughly and addresses how it can best be used.
- ▶ M. D. Svinicki, *Learning and Motivation in the Postsecondary Classroom* (San Francisco: Jossey Bass, 2004). The only motivation book focused on college and university teaching, this book addresses both motivation and learning theories, with clear examples for implementation.